Risk-Based Portfolio Construction on the EURO STOXX 50, 2004–2026

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Abstract

This paper compares seven long-only weighting rules applied to the point-in-time constituents of the EURO STOXX 50 between February 2004 and June 2026: equal weight, capitalisation weight, inverse volatility, minimum variance, equal risk contribution, most diversified portfolio and hierarchical risk parity. After transaction costs of 10 basis points per side, the risk-based rules lower volatility by up to 22% relative to the EURO STOXX 50 Gross Return index, and the minimum-variance portfolio reduces the maximum drawdown from 53.7% to 42.0%. No rule delivers a mean active return, a Sharpe ratio or a factor-adjusted alpha that can be told apart from zero. Relative performance is also unstable: the three most risk-focused rules outperformed the index in 2004–2019 and underperformed it by about 5.5% a year in 2020–2026. Regressions of the difference from the index on European factors attribute the lower risk to a smaller market beta and to tilts towards profitable and conservatively investing firms.

1 Introduction

Weighting rules that rely only on a covariance matrix are widely used to build equity portfolios with lower risk than a capitalisation-weighted index. They need no forecast of expected returns, and for this reason they are often presented as a more robust alternative to both market-cap weighting and mean-variance optimisation. Whether the reduction in risk comes at a cost in return, and whether any additional return can be distinguished from chance in a sample of realistic length, is an empirical question.

This paper examines it on a deliberately simple universe, the 50 stocks of the EURO STOXX 50. There is no stock selection and no return forecast: each strategy only decides how much to hold of each eligible constituent. The investigation is organised around one question. Do weighting rules that use only a covariance matrix improve on the EURO STOXX 50 itself, once transaction costs and estimation error are taken into account?

The analysis follows four principles. Index membership is point in time, so that stocks removed from the index remain in the history and the results carry no survivorship bias. Weights use only information available at the formation date, and a one-month execution lag is tested as a check. Transaction costs are charged from the start, and gross and net results are both reported. Finally, performance is measured against the EURO STOXX 50 Gross Return index, and factor attribution is applied to the difference between each strategy and the index, not to absolute returns.

The sample covers February 2004 to June 2026 and contains 269 monthly returns. Section 2 describes the data and Section 3 the seven strategies. Section 4 sets out the evaluation design.

Sections 5 to 8 present the results, the statistical tests, the factor attribution, and the analysis of subperiods and robustness. Sections 9 and 10 discuss the limitations and conclude.

2 Data

2.1 Index membership and identifiers

All market data are taken from LSEG Workspace. Index membership is rebuilt from two exports for the STOXX 50 index: the record of joiners and leavers, with 231 records from January 1998 to October 2026, and the list of the 50 current constituents. Eleven constituents appear in the joiners-and-leavers file under Frankfurt codes and are mapped to the corresponding Xetra codes used in the price data. The reconstructed membership counts 50 stocks at the first and at the last checkpoint, and the log of anomalies is empty.

Securities are identified by PermID through a crosswalk of 118 RICs mapped to 117 securities, so that changes of ticker, listing or share line do not interrupt a company's history. Three issuers have two RICs under the same PermID, and in each case the monthly price series of the two lines were compared with a guard requiring relative differences below 1%. The guard flagged distinct securities in all three groups. For Volkswagen (ordinary and preference shares) and Telecom Italia (an older and a current line) the median relative price differences are 13.5% and 74.6%, and the two lines are treated as separate securities. Unilever is kept as a single security. Its two lines were index members one after the other, `UNc.AS` from August 1998 to 1 July 2019 and `UNA.AS` from that date to 30 November 2020, and their monthly returns have a correlation of 0.96 over the 241 months in which both exist. Keeping them together preserves a continuous return history, which the eligibility rule below requires.

2.2 Panel, eligibility and exits

The resulting panel has 30,309 security-month observations. A stock is eligible on a formation date if it is an index member on that date and has a non-missing monthly return in each of the previous 60 months. A single gap in the window makes the stock ineligible until 60 complete months have accumulated again. At each of the 270 formation dates, between 46 and 50 stocks pass this rule.

Missing observations are classified, and no temporary gap inside a series occurs in the panel. There are 24 permanent exits. For a permanent exit the portfolio books a terminal return of zero in the last month. The table of documented terminal returns has a single entry, Sanpaolo IMI in December 2006, also at zero. Of the 24 exits, only Sanpaolo IMI was still an index member on the formation date of its last month, so the assumption affects one position in one month. A few market capitalisations are missing. In the capitalisation-weighted portfolio those stocks receive zero weight and the remaining weights are renormalised.

As a check on the panel, a capitalisation-weighted portfolio of the reconstructed universe was compared with the EURO STOXX 50 Gross Return index. Their monthly returns have a correlation of 0.9965, and the reconstructed portfolio trails the index by about 0.28 percentage points a year. The same shortfall appears when the 60-month history filter is not applied, so it is not caused by the filter.

3 Portfolio construction

Seven rules are compared. Two are naive references and five use risk estimates. All portfolios are long only, fully invested and rebalanced every month. Table 1 lists the rules.

Table 1. Weighting rules

Strategy	Rule	Covariance input
Equal weight	Weight $1/N$ for each eligible stock	None
Cap-weighted	Weight proportional to market capitalisation	None
Inverse volatility	Weight proportional to $1/\sigma_i$	Sample
Minimum variance	Long-only global minimum variance, cap of 5% per stock	Ledoit–Wolf
Equal risk contribution	Equal share of portfolio risk for each stock	Ledoit–Wolf
Most diversified	Maximum diversification ratio, cap of 5% per stock	Ledoit–Wolf
Hierarchical risk parity	Recursive bisection on a clustering tree, single linkage	Ledoit–Wolf

Let w be the vector of weights, Σ the covariance matrix of eligible stocks and σ the vector of their volatilities. The minimum-variance portfolio solves

$$\min_{w} \; w^{\top}\Sigma w \quad \text{subject to} \quad \mathbf{1}^{\top}w = 1, \;\; 0 \le w_i \le 0.05,$$

as a quadratic program. The most diversified portfolio maximises $w^{\top}\sigma / \sqrt{w^{\top}\Sigma w}$ under the same constraints, solved through a convex reformulation that allows the cap. The equal-risk-contribution portfolio equalises $w_i(\Sigma w)_i$ across stocks and is computed by cyclical coordinate descent. Hierarchical risk parity builds a tree of stocks from their correlations and splits the weight recursively between clusters in inverse proportion to their variance; the base case uses single linkage. With 46 to 50 candidates, equal weighting corresponds to about 2% per stock, so the 5% cap is binding only for concentrated solutions.

The covariance matrix is estimated on the previous 60 months of returns, and at least 20 stocks must have a complete window. Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) is applied for the four optimisation-based rules. Inverse volatility needs only variances, so the sample covariance is kept for it. The minimum-variance and most diversified portfolios follow Clarke et al. (2006) and Choueifaty and Coignard (2008), equal risk contribution follows Maillard et al. (2010), and hierarchical risk parity follows López de Prado (2016).

Weights formed at the end of month t earn the return of month $t+1$. After each month the weights drift with returns, and rebalancing back to target weights costs 10 bp per side on the traded weight. Stocks leaving the index follow the exit rules of Section 2.

4 Evaluation design

Benchmark and risk-free rate. The reference is the EURO STOXX 50 EUR Gross Return index (RIC `.SX5GT`), which reinvests gross dividends and is therefore comparable with the total-return series of the strategies. A price index would credit every strategy with the dividend yield as spurious alpha. Sharpe ratios use excess returns over the one-month Euribor from the ECB Data Portal, which averages 1.16% a year over the sample. A US Treasury bill series is available in the factor files, but it is in the wrong currency for a euro

investor and is kept only as a reference column.

Factors. Attribution uses the Fama–French Europe factors for market, size, value, profitability and investment (Fama and French, 2015) and the momentum factor (Carhart, 1997). They are expressed in US dollars and cover developed Europe, not the euro-area large caps. As a consequence the market beta of the index against these factors is attenuated, between 0.73 and 0.77, and absolute alphas look large, between 2.55% and 3.03% a year for the index itself. Absolute alphas are therefore not interpreted. Each strategy is regressed on the factors after subtracting the index return, so that the common mismatch largely cancels and a coefficient reads as the amount of a factor held in excess of the index.

Inference. Mean active returns and regression coefficients use Newey–West standard errors (Newey and West, 1987). Differences in Sharpe ratios and ratios of volatilities use a circular block bootstrap (Politis and Romano, 1992) with blocks of six months and 5,000 replications, which preserves serial dependence and the cross-correlation of the series; the approach is in the spirit of Ledoit and Wolf (2008). Seeds are fixed. Families of comparisons are adjusted with the Holm method (Holm, 1979).

Subperiods and robustness. Four subperiods are fixed in advance and follow economic episodes: 2004–2007, 2008–2009, 2010–2019 and 2020–2026. Robustness is assessed by changing one element at a time from the minimum-variance and hierarchical-risk-parity base cases, without re-selecting the base case from the variants.

5 Results: performance, risk and costs

Table 2 reports annualised results for the whole sample, net of 10 bp per side unless marked.

Table 2. Performance, February 2004 to June 2026 (269 months)

	Return (net)	Volatility	Sharpe	Max draw-down	Monthly turnover	Return (gross)
Equal weight	7.45%	17.36%	0.43	−54.59%	2.79%	7.52%
Cap-weighted	7.16%	16.22%	0.44	−51.88%	1.07%	7.19%
Inverse volatility	7.70%	16.21%	0.47	−52.98%	2.75%	7.77%
Minimum variance	7.71%	12.95%	0.55	−42.03%	5.23%	7.85%
Equal risk contribution	7.09%	15.52%	0.44	−52.44%	3.07%	7.16%
Most diversified	7.17%	13.66%	0.49	−44.19%	5.81%	7.31%
Hierarchical risk parity	7.22%	14.64%	0.47	−48.76%	10.17%	7.48%
EURO STOXX 50 Gross Return	7.45%	16.54%	0.45	−53.69%	–	–

Returns and volatility are annualised. Sharpe ratios use excess returns over the one-month Euribor. Turnover is the average monthly fraction of the portfolio traded.

Net returns lie in a narrow range, from 7.09% to 7.71% a year, while volatility ranges from 12.95% to 17.36%. Minimum variance matches the return of the index with 22% less volatility, and most diversified gives up 28 bp of return for 17% less volatility. The maximum

drawdown follows the order of volatility, from −42.03% for minimum variance to −54.59% for equal weight, against −53.69% for the index.

Costs scale with turnover. The difference between gross and net return is 3 bp for the capitalisation-weighted portfolio, 7 bp for equal weight, inverse volatility and equal risk contribution, 14 bp for minimum variance and most diversified, and 26 bp for hierarchical risk parity, whose average monthly turnover is 10.17%. Hierarchical risk parity moves from 3 bp ahead of the index before costs to 23 bp behind it after costs. Tracking error rises with the reduction in risk: 1.40% for the capitalisation-weighted portfolio, between 2.42% and 2.88% for inverse volatility, equal weight and equal risk contribution, 4.18% for hierarchical risk parity, 5.92% for most diversified and 6.83% for minimum variance. Information ratios lie between −0.24 and 0.07.

Figure 1 shows the growth of 100 invested in four of the strategies and in the index. The lead of minimum variance and hierarchical risk parity builds mainly between 2009 and 2020 and narrows afterwards. Drawdowns move together across strategies (Figure 2), and minimum variance is shallower in 2008–2009, with a maximum drawdown of −34.5% against −46.8% for the index. In calendar years (Figure 3), minimum variance loses less than the other strategies in 2008 and trails clearly in the rebound of 2009.

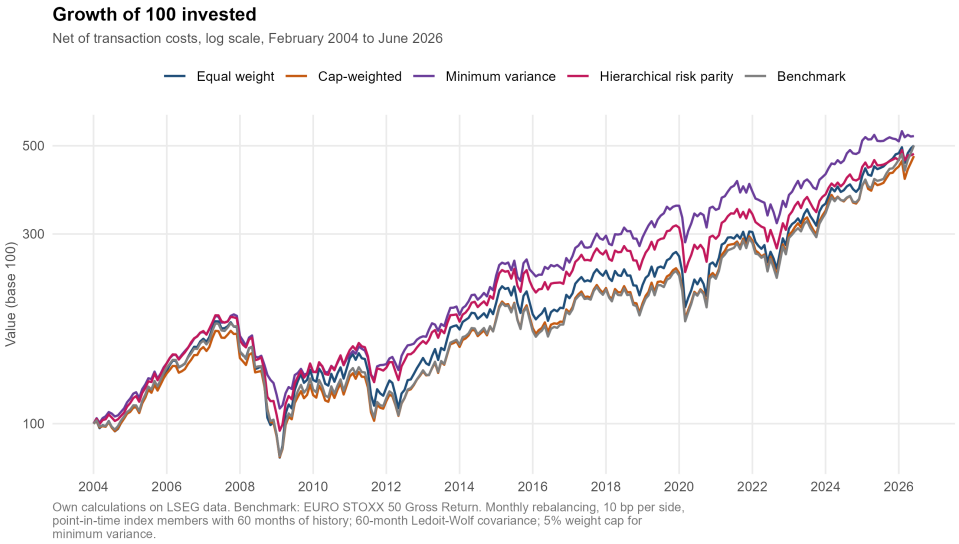


Figure 1. Growth of 100 invested, net of transaction costs (log scale)

Figure 4 plots return against volatility for all seven strategies. Minimum variance has the lowest volatility and the highest return, with inverse volatility a close second on return. Equal weight and the index have the same return and the highest risk. Figure 5 shows monthly turnover: hierarchical risk parity renews about a tenth of the portfolio every month without quiet periods, the capitalisation-weighted portfolio stays near zero with occasional spikes, and minimum variance, most diversified and hierarchical risk parity all show single months above 20%.

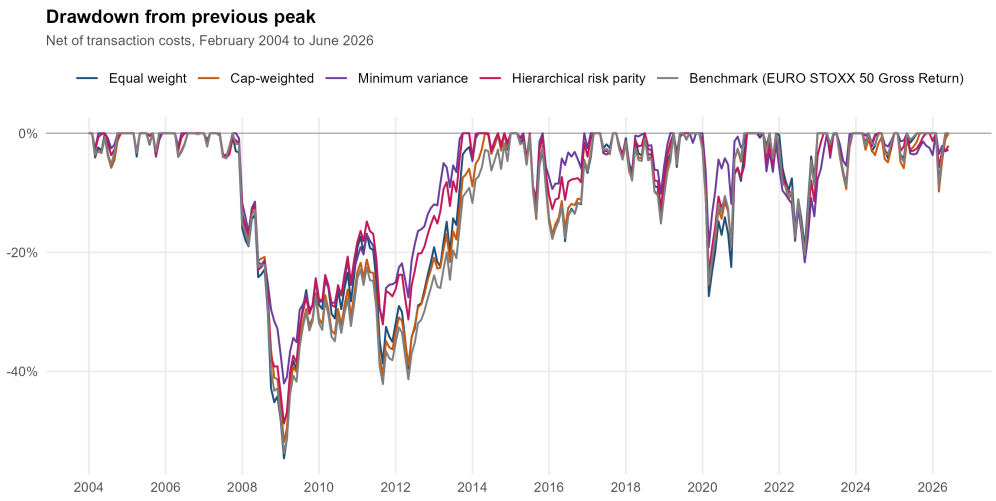


Figure 2. Drawdown from the previous peak, net of transaction costs



Figure 3. Calendar-year returns, net of transaction costs

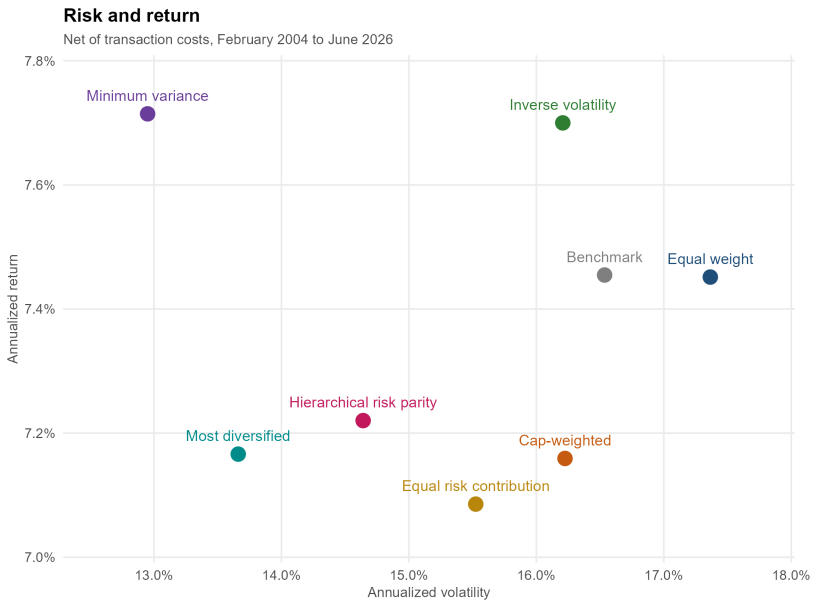


Figure 4. Return and volatility of the seven strategies, net of transaction costs

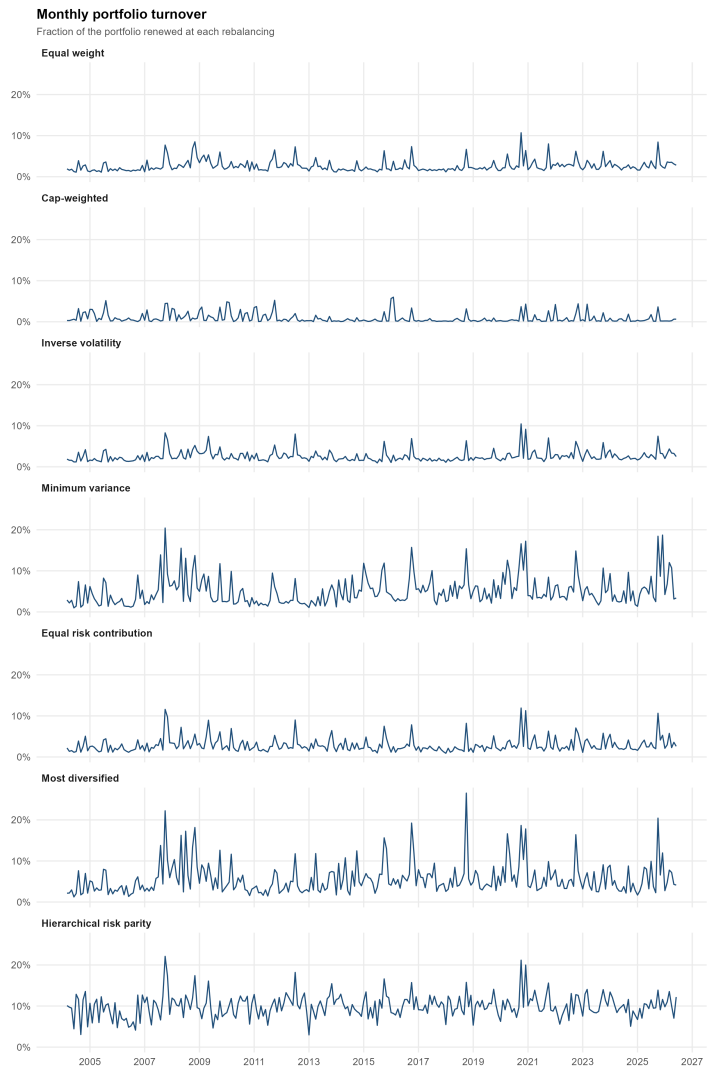


Figure 5. Monthly portfolio turnover by strategy

6 Statistical tests

Table 3 tests the mean active return and the market model against the index, and Table 4 tests the Sharpe ratio and the volatility. All use net returns.

Table 3. Active return and market model versus the index, net

	Mean active return	p-value	Market beta	Alpha	p-value
Equal weight	0.14%	0.797	1.038	−0.15%	0.766
Cap-weighted	−0.33%	0.284	0.978	−0.17%	0.573
Inverse volatility	0.18%	0.725	0.970	0.40%	0.391
Minimum variance	−0.29%	0.854	0.722	1.77%	0.092
Equal risk contribution	−0.51%	0.439	0.926	0.04%	0.940
Most diversified	−0.71%	0.599	0.778	0.94%	0.319
Hierarchical risk parity	−0.52%	0.591	0.861	0.52%	0.486

Mean active returns and alphas are annualised. Standard errors are Newey–West.

Table 4. Sharpe and volatility tests versus the index, net

	Sharpe difference	95% interval	p-value	Volatility ratio	95% interval	p-value (Holm)
Equal weight	−0.013	[−0.071, 0.044]	0.647	1.050	[1.018, 1.083]	0.004
Cap-weighted	−0.012	[−0.049, 0.025]	0.526	0.981	[0.967, 0.996]	0.028
Inverse volatility	0.020	[−0.037, 0.077]	0.501	0.980	[0.956, 1.005]	0.113
Minimum variance	0.101	[−0.063, 0.266]	0.230	0.783	[0.733, 0.837]	0.000
Equal risk contribution	−0.004	[−0.074, 0.066]	0.915	0.939	[0.916, 0.962]	0.000
Most diversified	0.042	[−0.094, 0.179]	0.552	0.826	[0.774, 0.882]	0.000
Hierarchical risk parity	0.023	[−0.078, 0.123]	0.654	0.885	[0.858, 0.914]	0.000

Circular block bootstrap, blocks of six months, 5,000 replications. The p-values for the Sharpe differences are not adjusted.

Mean active returns are small and none is significant, with p-values between 0.28 and 0.85. The capitalisation-weighted shortfall of 33 bp a year is the closest to a pattern, and it is in line with the 0.28 percentage point shortfall of the reconstructed index in Section 2.

No Sharpe improvement can be proved. All differences against the index have intervals that include zero, and after the Holm adjustment the smallest p-value among the pairwise Sharpe comparisons is 0.645. The largest estimate, 0.10 for minimum variance, has a bootstrap standard error of 0.084, so that only differences above roughly 0.17 would have been detectable. Differences among the strategies are also insignificant. Net of costs, minimum variance minus equal weight is −0.43% a year ($p = 0.804$) and minimum variance minus the capitalisation-weighted portfolio is 0.04% ($p = 0.978$).

Volatility is reduced significantly. Minimum variance, most diversified, hierarchical risk parity and equal risk contribution have volatility ratios below one with Holm-adjusted p-values below 0.001. Minimum variance runs at 0.78 times the volatility of the index, and equal weight at 1.05 times, which is also significant. Inverse volatility is statistically close to the index (0.98, Holm $p = 0.113$). Lower betas accompany the reduction in volatility: market

betas are between 0.97 and 1.04 for the naive rules and fall to 0.72 for minimum variance, 0.78 for most diversified and 0.86 for hierarchical risk parity. The market-model alpha of minimum variance, 1.77%, is the only one near conventional significance ($p = 0.092$), and with seven strategies tested it would not be convincing even at that level.

The evidence supports the statement that these rules deliver lower risk for the same return. It does not support the statement that they deliver a higher risk-adjusted return.

7 Factor attribution relative to the index

Table 5 reports the six-factor regressions of each strategy's return minus the index return. The first row gives the loadings of the index itself in levels, as a reference.

Table 5. Relative loadings on the six factors, net

	Alpha	MKT	SMB	HML	RMW	CMA	MOM	R^2
EURO STOXX 50 Gross Return (levels)	2.55%	0.725***	−0.606***	0.244**	0.070	−0.375***	−0.105**	0.812
Equal weight	0.78%	0.003	0.064**	0.087**	−0.005	0.045	−0.087***	0.394
Cap-weighted	−0.37%	−0.012	−0.016	−0.025	0.077***	0.032	−0.006	0.212
Inverse volatility	0.66%	−0.022**	0.068**	0.003	0.108**	0.160***	−0.071***	0.181
Minimum variance	−0.29%	−0.132***	0.054	−0.253***	0.332***	0.452***	0.044	0.589
Equal risk contribution	−0.26%	−0.047***	0.050	−0.018	0.150***	0.132*	−0.034	0.204
Most diversified	−1.05%	−0.106***	0.079	−0.149**	0.339***	0.304***	0.039	0.482
Hierarchical risk parity	−0.11%	−0.062***	0.082**	−0.156***	0.226***	0.325***	−0.038	0.402

Alpha is annualised. Newey–West standard errors. *, ** and *** denote significance at 10%, 5% and 1%.

The index is a large-cap portfolio (SMB −0.61), with a negative investment loading (CMA −0.38), a small value tilt (HML 0.24) and a small negative momentum loading. Its alpha of 2.55% is not significant and, as explained in Section 4, largely reflects the fit of a euro-area index to dollar-denominated pan-European factors.

Minimum variance, most diversified and hierarchical risk parity share a common profile. Relative to the index they hold less market risk (MKT between −0.06 and −0.13), have a lower value or higher growth orientation (HML between −0.15 and −0.25), and hold more profitable firms (RMW between 0.23 and 0.34) and more conservatively investing firms (CMA between 0.30 and 0.45). This is the profile usually associated with low-volatility portfolios. The model explains 40% to 59% of their difference from the index. Inverse volatility and equal risk contribution show the same signs with lower intensity (RMW between 0.11 and 0.15, CMA between 0.13 and 0.16), and the model explains 18% to 20%. Equal weight is the only rule that tilts towards size and value (SMB 0.064, HML 0.087) and away from momentum (MOM −0.087), as expected when every constituent receives the same weight. The capitalisation-weighted portfolio differs from the index only through a small profitability loading.

No relative alpha is significant. In the six-factor model the alphas range from −1.05% to 0.78% a year and all have p-values above 0.17. The four-factor model with market, size, value and momentum gives the same answer, with relative alphas between −0.10% and 0.98%;

inverse volatility is the closest to significance (0.98%, $p = 0.053$) and minimum variance has 0.69% ($p = 0.514$). The risk-based rules did not add return beyond the factors they tilt towards. What they changed is the exposure, towards less market beta and more quality.

8 Stability: subperiods and robustness

8.1 Subperiods

Table 6 reports the geometric active return against the index in each subperiod, and Table 7 compares minimum variance with the index in more detail.

Table 6. Active return versus the index by subperiod, net (geometric, a year)

	2004–2007	2008–2009	2010–2019	2020–2026
Equal weight	−0.13%	2.36%	0.65%	−1.90%
Cap-weighted	−1.28%	0.26%	0.58%	−1.34%
Inverse volatility	1.21%	1.96%	1.22%	−2.53%
Minimum variance	1.78%	1.10%	3.17%	−5.45%
Equal risk contribution	1.60%	−1.26%	1.29%	−3.76%
Most diversified	1.08%	1.09%	2.33%	−5.68%
Hierarchical risk parity	1.36%	1.75%	1.91%	−5.26%

Table 7. Minimum variance and the benchmark by subperiod, net

Subperiod		Return	Volatility	Sharpe	Max draw-down
2004–2007 (47 months)	Minimum variance	17.28%	7.63%	1.76	−3.97%
	Benchmark	15.50%	9.11%	1.32	−5.21%
2008–2009 (24 months)	Minimum variance	−12.92%	18.24%	−0.80	−34.51%
	Benchmark	−14.01%	26.94%	−0.52	−46.83%
2010–2019 (120 months)	Minimum variance	9.58%	12.27%	0.80	−17.34%
	Benchmark	6.41%	15.15%	0.48	−25.33%
2020–2026 (78 months)	Minimum variance	6.40%	14.22%	0.41	−21.67%
	Benchmark	11.85%	17.79%	0.64	−23.25%

In each of the first three regimes, minimum variance, most diversified and hierarchical risk parity beat the index, with the largest gain for minimum variance in 2010–2019 (3.17% a year and a Sharpe ratio of 0.80 against 0.48). From 2020 the sign reverses for all seven strategies, and most sharply for the three most risk-focused ones, at −5.3% to −5.7% a year. Volatility remains lower than that of the index in this period (14.22% for minimum variance against 17.79%), so the shortfall concerns return and not risk control. The rolling 36-month active return in Figure 6 reaches about −7% for minimum variance and hierarchical risk parity at June 2026.

The full-sample result is therefore an average of two different periods. The data cannot say whether 2020–2026 is a temporary phase of strong large-cap and growth leadership or the end of a premium. Both readings are consistent with the factor profile of Section 7.

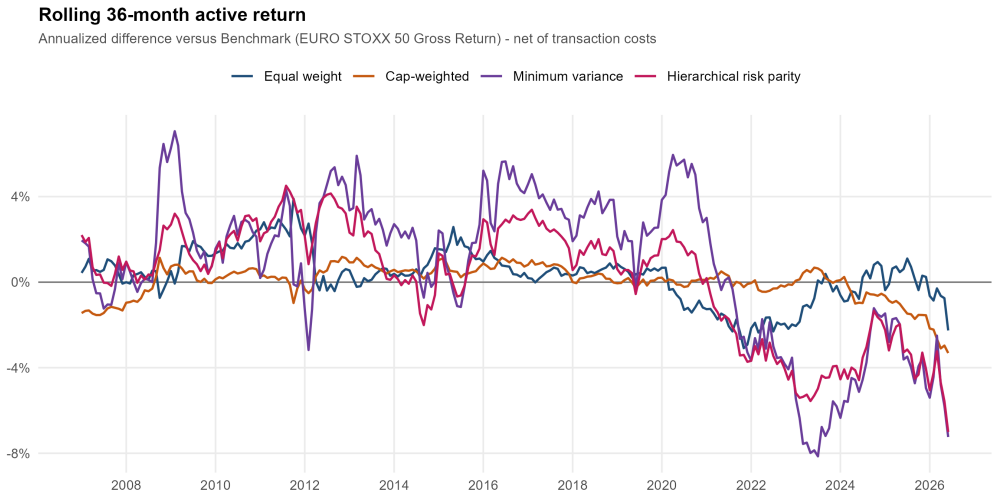


Figure 6. Rolling 36-month active return versus the index, net of transaction costs

8.2 Robustness

Table 8 re-estimates the minimum-variance and hierarchical-risk-parity strategies after changing one element at a time.

Table 8. Robustness of minimum variance and hierarchical risk parity, net

	Return	Volatility	Sharpe	Active return	Monthly turnover
Minimum variance, base case	7.71%	12.95%	0.55	0.26%	5.23%
Estimation window of 36 months	7.72%	13.32%	0.54	0.26%	7.44%
Sample covariance instead of shrinkage	7.65%	12.97%	0.54	0.20%	5.40%
Weight cap of 3%	7.50%	14.55%	0.49	0.05%	4.16%
Weight cap of 10%	7.07%	12.91%	0.50	−0.39%	7.03%
No weight cap	6.68%	13.02%	0.47	−0.78%	7.77%
Quarterly rebalancing	8.02%	12.98%	0.57	0.56%	3.41%
Execution lag of one month	7.81%	13.06%	0.55	0.42%	5.36%
Costs of 5 bp per side	7.78%	12.95%	0.55	0.33%	5.23%
Costs of 25 bp per side	7.51%	12.95%	0.54	0.06%	5.23%
Costs of 50 bp per side	7.18%	12.95%	0.51	−0.28%	5.23%
Hierarchical risk parity, single linkage (base)	7.22%	14.64%	0.47	−0.23%	10.17%
Hierarchical risk parity, average linkage	7.34%	14.78%	0.48	−0.12%	11.10%
Hierarchical risk parity, Ward linkage	6.97%	14.83%	0.45	−0.49%	11.34%

Active return is geometric and annualised.

The estimation choices matter little. A 36-month window or the plain sample covariance changes the return by at most 6 bp and the Sharpe ratio by 0.01, although the shorter window raises turnover from 5.23% to 7.44%. The weight cap matters more. Volatility stays near 13% for caps of 5%, 10% and no cap, but the return falls from 7.71% to 6.68% and turnover rises from 5.23% to 7.77% as the cap is loosened. A tighter cap of 3% raises volatility to 14.55% and lowers the Sharpe ratio to 0.49. The 5% cap gives the highest Sharpe ratio of the four settings, but the differences are modest and the cap is better read as a sensible choice than as

an optimum.

Quarterly rebalancing would have done better for minimum variance: the return rises to 8.02%, the Sharpe ratio to 0.57 and turnover falls by about one third to 3.41%. The base case keeps monthly rebalancing for every strategy so that the comparison is uniform, which means the minimum-variance results in Section 5 are slightly conservative. An execution lag of one month gives 7.81%, so the result does not depend on trading at formation-date prices. At 50 bp per side, five times the base case, minimum variance keeps a Sharpe ratio of 0.51 against 0.45 for the index, with a return of 7.18%, 27 bp below the index. Finally, the choice of linkage changes the return of hierarchical risk parity by up to 37 bp, from 6.97% to 7.34%, with turnover between 10% and 11%, and in none of the three versions does it match the 7.45% of the index.

9 Limitations

Statistical power is limited. With 269 monthly observations, the bootstrap standard error of a Sharpe-ratio difference is 0.084 for minimum variance, so a true improvement of 0.10 would be missed more often than not. A result that is not significant should be read as not proven, not as absent.

The universe is narrow. The study uses 46 to 50 large caps at each date, and the dispersion of risk across such stocks is limited. The gains from covariance-based weighting could differ in a broader universe, such as a mid-cap or pan-European index.

The factor data do not match the market. The Fama–French Europe factors are in US dollars and cover a wider region than the euro-area large caps. The relative attribution addresses the main distortion, but some noise remains in the loadings, and a variant built on euro style indices would be a cleaner test.

Design choices were fixed once. The 60-month window, the 5% cap, monthly rebalancing and the Ledoit–Wolf estimator are fixed in the base case and then varied one at a time, and their joint variation is not explored. The quarterly result shows that the base case is not the best setting for minimum variance, so the headline numbers are not an optimised outcome, but neither do they represent the full range. The four subperiods follow economic episodes and not a statistical break test, and the reversal after 2019 rests on 78 months and is not tested for significance.

The cost model is simple: a flat 10 bp per side on traded weight at month-end prices, without market impact, time-varying spreads, taxes or capacity limits. Hierarchical risk parity, with monthly turnover of about 10%, is the most exposed to this simplification.

Finally, some data treatments affect the results. The 60-month history filter removes the youngest stocks from every strategy. Every permanent exit books a terminal return of zero, which could differ from the true outcome for a stock that left after a takeover or a failure, although only one of the 24 exits was an index member on its last formation date. The reconstructed capitalisation-weighted portfolio trails the official index by about 0.28 percentage points a year. The input data come from LSEG Workspace under licence and cannot be published with the code.

10 Conclusion

Over February 2004 to June 2026, risk-based weighting of the EURO STOXX 50 constituents lowered volatility and drawdowns by amounts that are statistically solid. Minimum variance cut volatility by 22% and the worst drawdown by 11.7 percentage points at a trading cost of 14 bp a year, and the result is stable to the covariance estimator, the estimation window and the level of costs. These gains came with market betas between 0.72 and 0.86 for the three most risk-focused rules. The data do not show a gain in return or in the Sharpe ratio, either against the index or against equal weighting: the best Sharpe estimate, 0.55 for minimum variance against 0.45 for the index, comes with a difference whose interval runs from -0.06 to $+0.27$.

The advantage also depends on the period. The three most risk-focused rules outperformed clearly until 2019 and lost about 5.5% a year in 2020–2026. The data show the reversal without explaining it, and a user of these rules should expect long stretches of underperformance relative to the index, with tracking error between 4% and 7%. These strategies therefore suit a mandate that values lower risk and smaller drawdowns more than closeness to the index, and for minimum variance a lower-turnover version looks preferable in this sample. They do not suit a mandate that expects to beat the index through the weighting scheme alone.

Natural extensions are the replacement of the dollar-denominated pan-European factors with euro style indices, to test the quality and low-beta interpretation directly, the application of the same pipeline to a broader universe, a pre-specified out-of-sample comparison of monthly and quarterly rebalancing together with a turnover-penalised version of minimum variance, and a formal test of the 2020 reversal.

A Decision log

Table 9 lists the choices that affect the numbers and, where one was considered, the alternative that was not adopted.

Table 9. Methodological decisions

Topic	Decision	Alternative considered
Universe	Point-in-time EURO STOXX 50 members with at least 60 complete months of returns	–
Identifiers	Volkswagen and Telecom Italia split into two securities each; Unilever kept as one, on the evidence of consecutive membership and a return correlation of 0.96	–
Event codes	Eleven Frankfurt codes mapped to Xetra codes	–
Exits	Terminal return of zero in the last month of a permanent exit; Sanpaolo IMI documented explicitly in December 2006	–
Benchmark	EURO STOXX 50 EUR Gross Return (`.SX5GT`)	Price index, which credits the dividend yield as alpha
Risk-free rate	One-month Euribor from the ECB Data Portal, averaging 1.16% a year	US Treasury bill series from the factor file

Topic	Decision	Alternative considered
Factors	Fama–French Europe, applied to strategy minus index	Absolute attribution, which gives artefact alphas
Covariance	60 months; Ledoit–Wolf for minimum variance, equal risk contribution, most diversified and hierarchical risk parity; sample covariance for inverse volatility	Sample covariance throughout; 36-month window (Table 8)
Weight cap	5% for minimum variance and most diversified	3%, 10% and no cap (Table 8)
Cap-weighted portfolio	Missing market capitalisation set to zero weight, weights renormalised	–
Rebalancing and costs	Monthly, 10 bp per side on traded weight	Quarterly; 5, 25 and 50 bp (Table 8)
Hierarchical risk parity	Single linkage	Average and Ward linkage (Table 8)
Subperiods	2004–2007, 2008–2009, 2010–2019, 2020–2026, fixed in advance	–
Inference	Newey–West errors; circular block bootstrap with blocks of six months and 5,000 replications; Holm adjustment within each family	Unadjusted p-values

B Reproducibility

The analysis is written in R and organised as a main script and six modules covering the data layer, the point-in-time backtest engine, the portfolio constructors and risk estimators, evaluation, tables and figures, and final controls. Tables are written to CSV files and to one Excel workbook, and figures to PNG files. The Fama–French and ECB inputs are public. The LSEG inputs are licensed, so the data folders are not part of the public repository, and the data layer can be rebuilt only with LSEG Workspace access.

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